

Lab 2: Data Carving

What You Need for this lab

- Install Virtualbox : <https://www.virtualbox.org/wiki/Downloads>
- Install Kali 2021.4. : <https://old.kali.org/kali-images/kali-2021.4/>
 - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
- Run a tool installation script instructions, or you can simply follow the commands below : (the script ONLY is tested on Kali 2021.4)
 - `wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Help/tool-install-zsh.sh`
 - `chmod +x tool-install-zsh.sh`
 - `./tool-install-zsh.sh`

Example scenarios

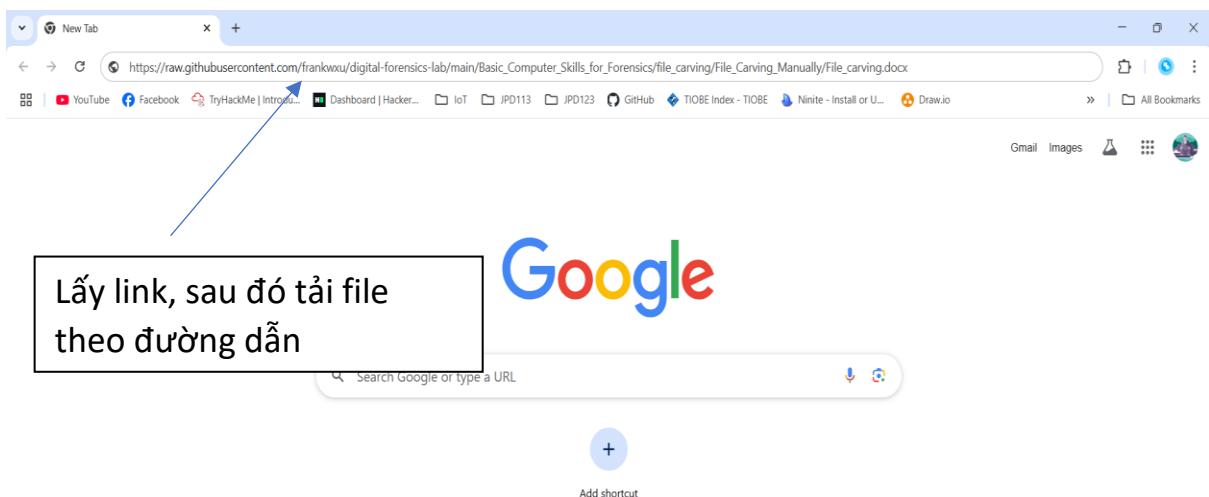
- Scenario 1: A file (A) is hidden inside of another file (B). You can't open the file B because the B's header is corrupted.
- Scenario 2: A suspect deleted files. The files contains an important information. A file occupies a few clusters. Unfortunately, some clusters are reused (overwritten) by new files.

A forensic expert really wants to recover files, even a partial files.

1. Extracting images from a corrupted Word document

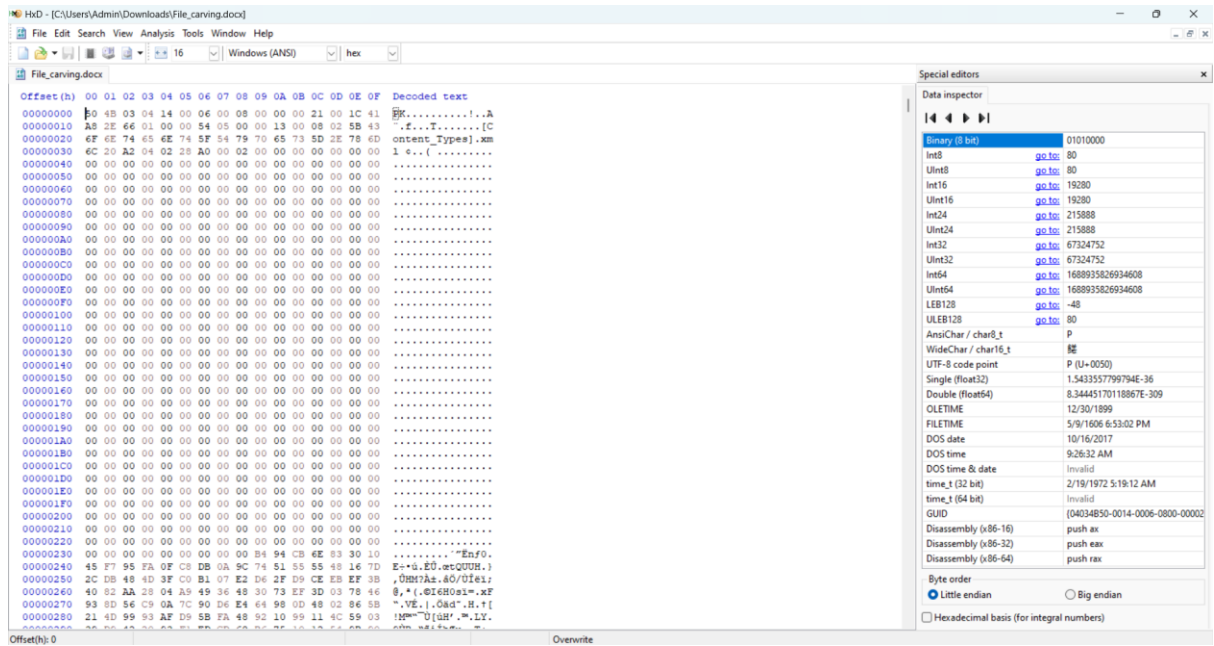
Step 1.

- Prepare required files



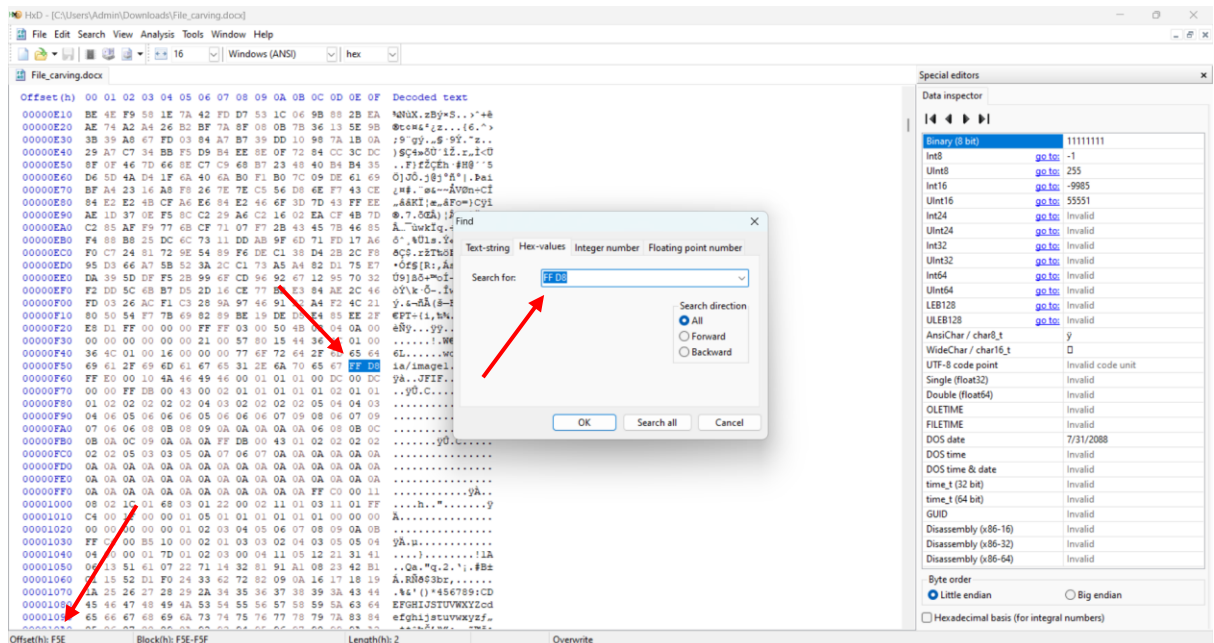
Step 2.

- View the file in a hex editor

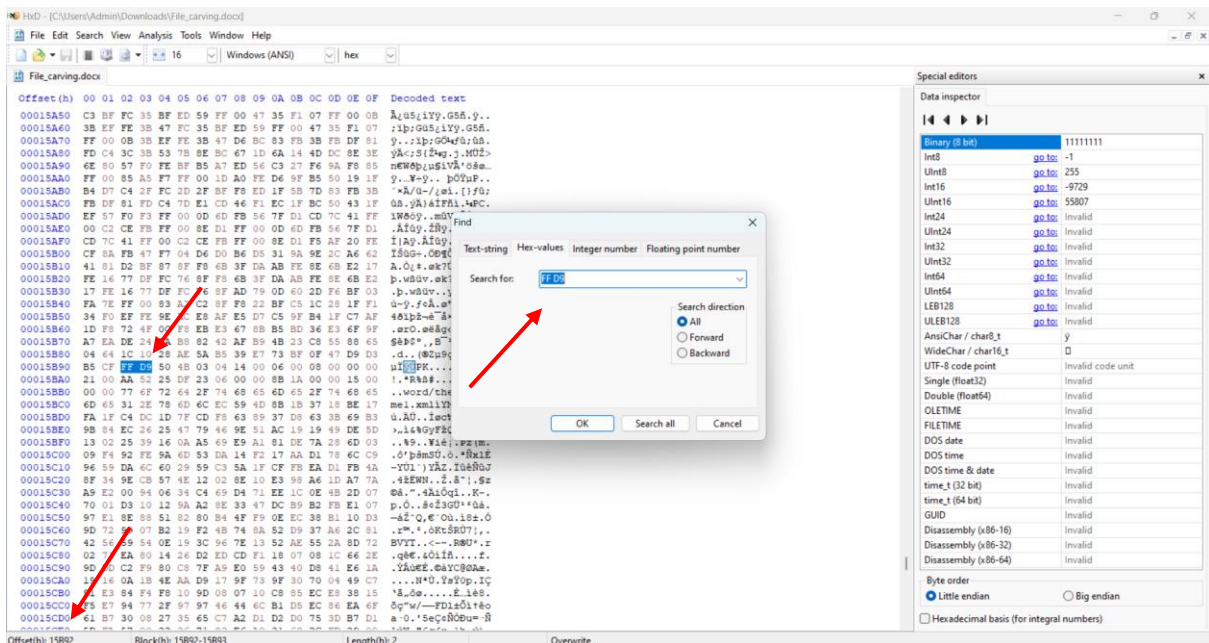


Step 3.

- Search file header start offset – 0F5E

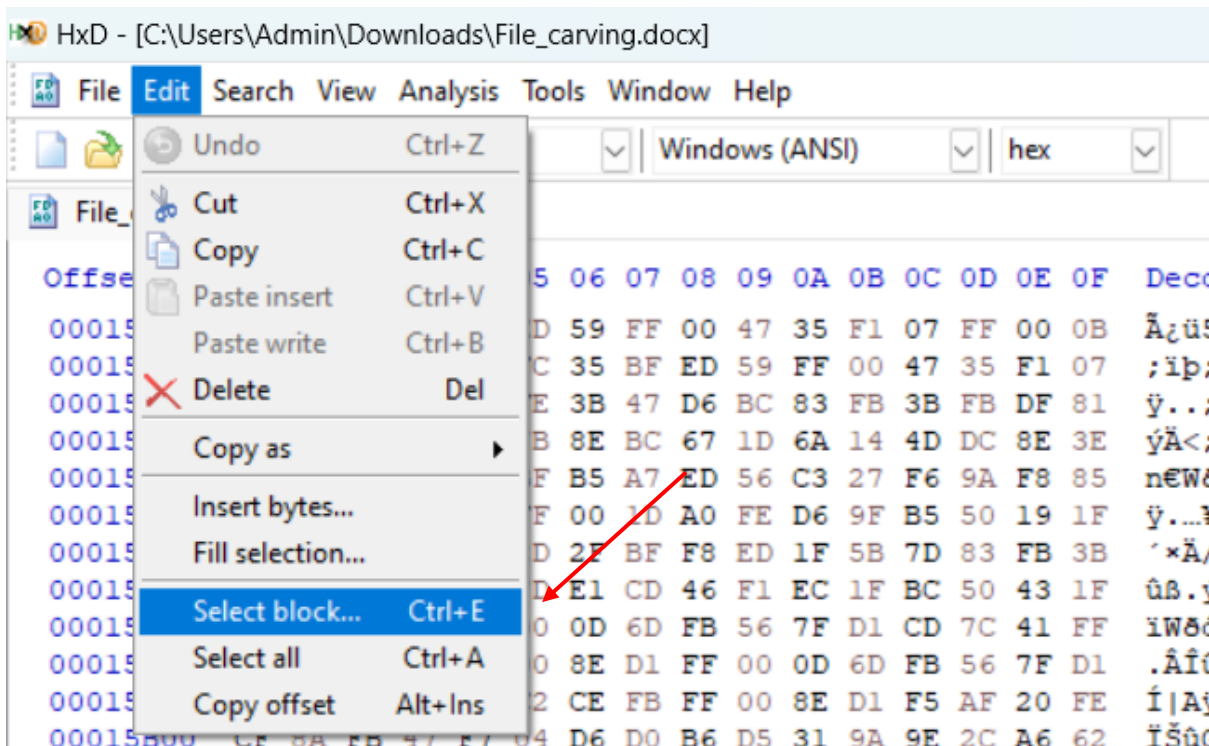


- Search file trailer ends offset – 15B93

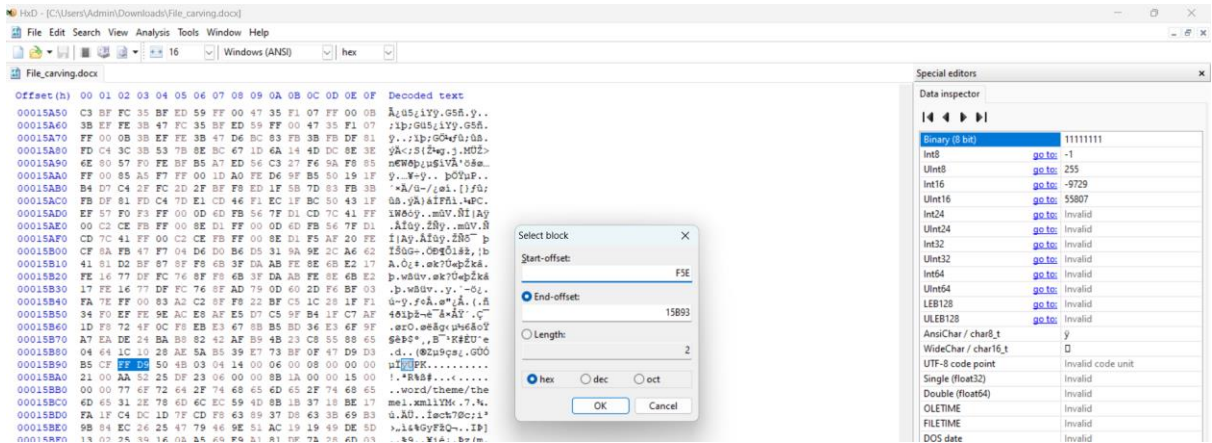


- Select hex from header to tail

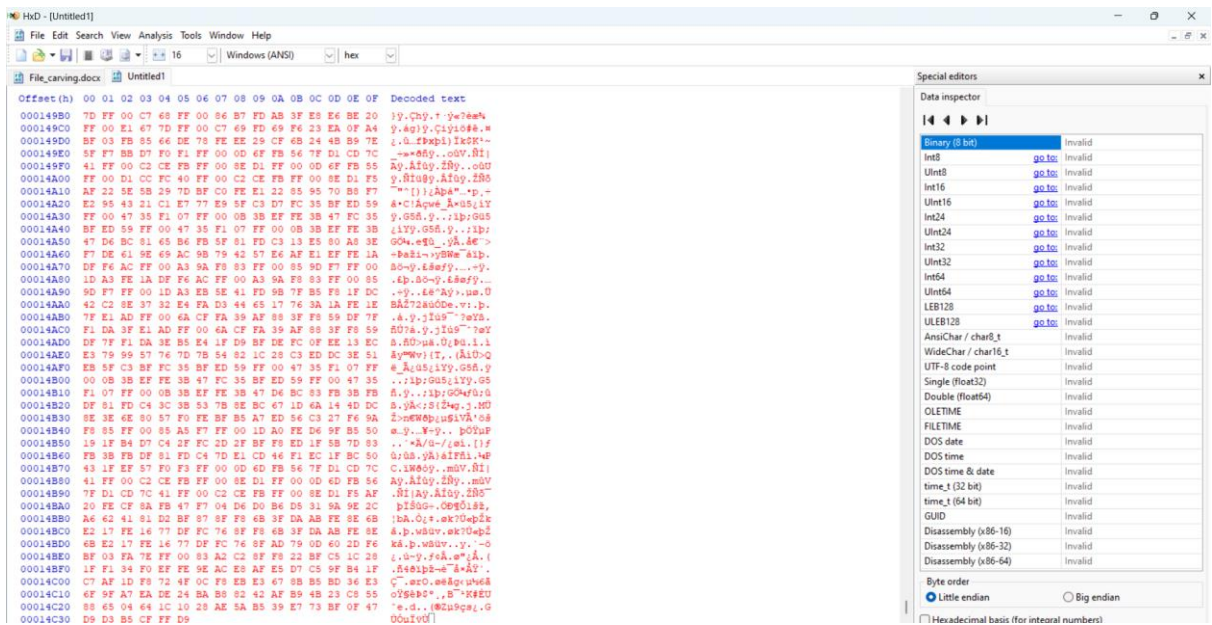
- $(0F5E)_{16} = (3934)_{10}$
- $(15B93)_{16} = (88979)_{10}$



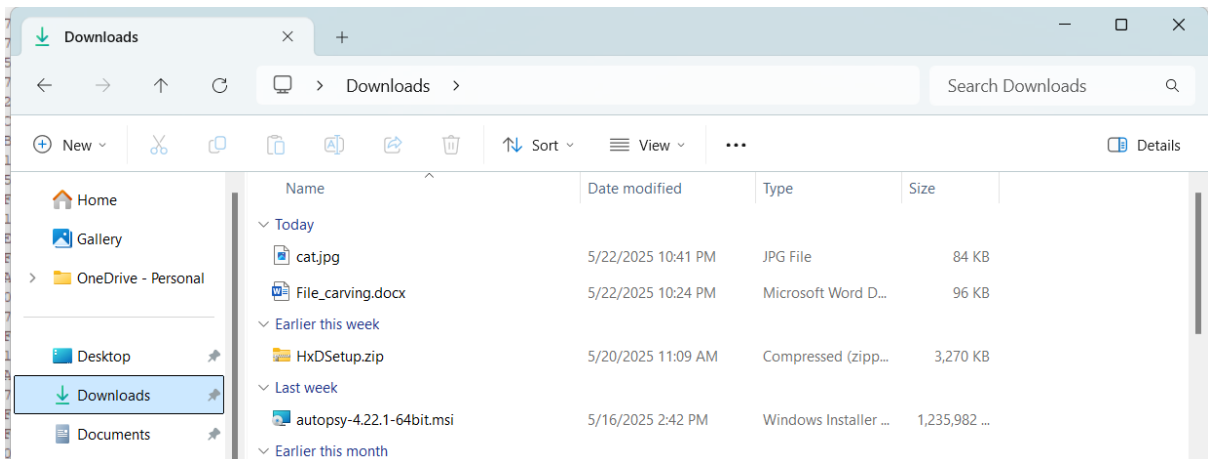
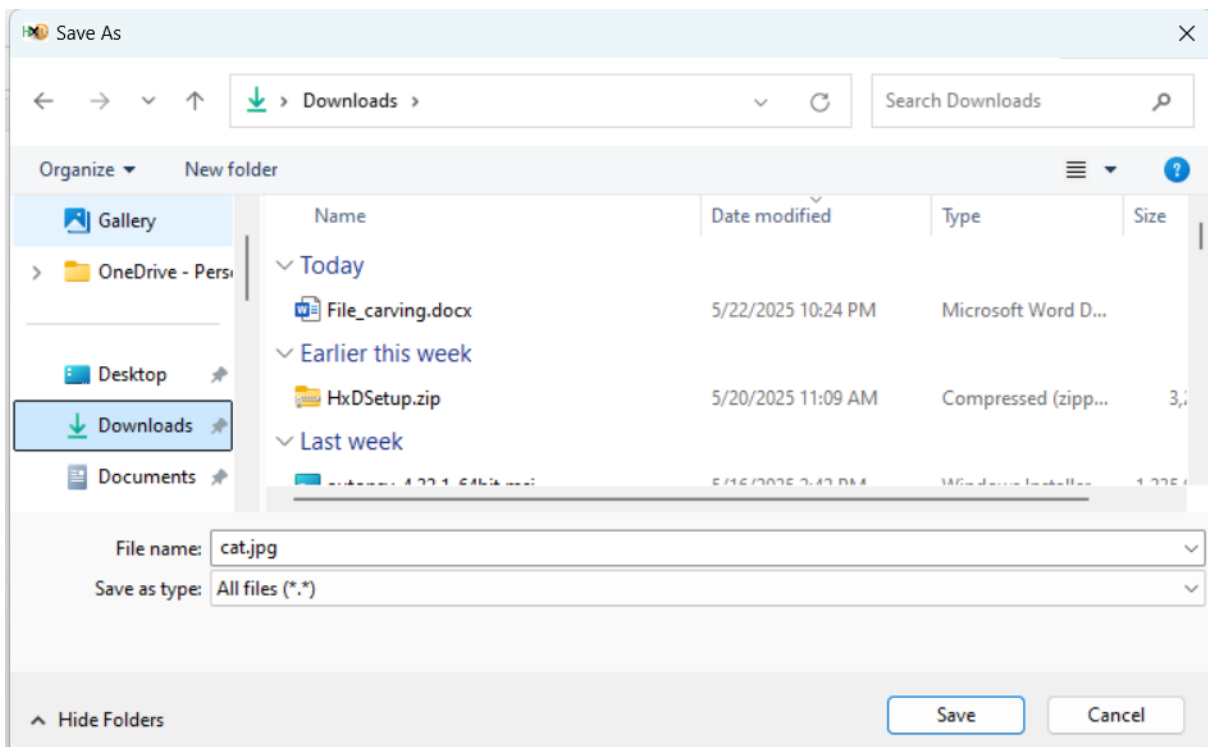
- Copy the selection



– Paste the selection

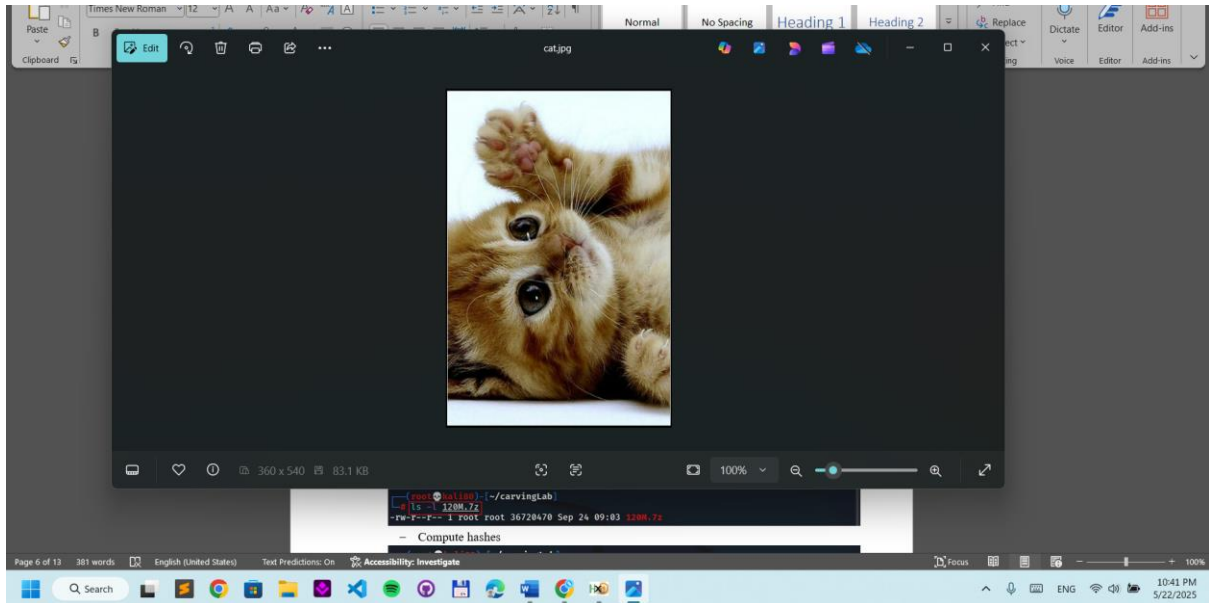


- Save the image



Step 4.

- Show the carved image



2. Carving/Recovering a USB image

- Prepare a USB image for file carving

Step 1.

- Download the zipped USB image

```
(hainguyen@hainguyen)-[~]
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z

(hainguyen@hainguyen)-[~]
$ ls -l 120M.7z
-rw-rw-r-- 1 hainguyen hainguyen 36720470 May 22 11:54 120M.7z
```

- Compute hashes

```
(hainguyen@hainguyen)-[~]
$ hashdeep -c md5,sha1 120M.7z
%%%% HASHDEEP-1.0
%%%% size,md5,sha1,filename
## Invoked from: /home/hainguyen
## $ hashdeep -c md5,sha1 120M.7z
##
36720470,dfe7b5424e54cd1bf50d5df47aceeb3c,2810745018afaa2da31dc17a8eb590fca6
6eeef7,/home/hainguyen/120M.7z

(hainguyen@hainguyen)-[~]
$ |
```

- List the content of the zipped file

```
(hainguyen@hainguyen)-[~]  
$ 7z l 120M.7z
```

```
7-Zip 24.08 (x64) : Copyright (c) 1999-2024 Igor Pavlov : 2024-08-11  
64-bit locale=en_US.UTF-8 Threads:1 OPEN_MAX:1024
```

```
Scanning the drive for archives:  
1 file, 36720470 bytes (36 MiB)
```

```
Listing archive: 120M.7z
```

```
--
```

```
Path = 120M.7z  
Type = 7z  
Physical Size = 36720470  
Headers Size = 214  
Method = LZMA2:24  
Solid = +  
Blocks = 1
```

Date	Time	Attr	Size	Compressed	Name
2021-09-22	22:59:31	D...A	0	0	120M
2021-09-22	22:59:31	A	124780544	36720256	120M/usb_fat_carving.001
2021-09-22	22:59:32	A	1685		120M/usb_fat_carving.001.txt
2021-09-22	22:59:32		124782229	36720256	2 files, 1 folders

```
(hainguyen@hainguyen)-[~]  
$ |
```

- List the content of the zipped file

```
(hainguyen@hainguyen)-[~]  
$ 7z e 120M.7z
```

```
7-Zip 24.08 (x64) : Copyright (c) 1999-2024 Igor Pavlov : 2024-08-11  
64-bit locale=en_US.UTF-8 Threads:1 OPEN_MAX:1024
```

```
Scanning the drive for archives:  
1 file, 36720470 bytes (36 MiB)
```

```
Extracting archive: 120M.7z
```

```
--
```

```
Path = 120M.7z  
Type = 7z  
Physical Size = 36720470  
Headers Size = 214  
Method = LZMA2:24  
Solid = +  
Blocks = 1
```

```
Everything is Ok
```

```
Folders: 1  
Files: 2  
Size: 124782229  
Compressed: 36720470
```

- Verify the hashes

```
(hainguyen@hainguyen)-[~]
$ hashdeep -c md5,sha1 usb_fat_carving.001
%%% HASHDEEP-1.0
%%% size,md5,sha1,filename
## Invoked from: /home/hainguyen
## $ hashdeep -c md5,sha1 usb_fat_carving.001
##
124780544,ba4a1d0ba49f4a6667b00a3b3e85e604,bcc2d49fd49c9521ecb1739f6542c6bf3
27375ef,/home/hainguyen/usb_fat_carving.001

(hainguyen@hainguyen)-[~]
$ cat usb_fat_carving.001.txt | grep "checksum"
MD5 checksum:      ba4a1d0ba49f4a6667b00a3b3e85e604
SHA1 checksum:     bcc2d49fd49c9521ecb1739f6542c6bf327375ef
MD5 checksum:      ba4a1d0ba49f4a6667b00a3b3e85e604 : verified
SHA1 checksum:     bcc2d49fd49c9521ecb1739f6542c6bf327375ef : verified

(hainguyen@hainguyen)-[~]
$ |
```

Step 2.

- Exam the content of the USB
- Display partitions

```
(hainguyen@hainguyen)-[~]
$ fdisk -l usb_fat_carving.001
Disk usb_fat_carving.001: 119 MiB, 124780544 bytes, 243712 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa1159a00

Device                Boot Start    End Sectors  Size Id Type
usb_fat_carving.001p1 *        128 243711   243584 118.9M  e W95 FAT16 (LBA)
```


- Find deleted files

```
(hainguyen@hainguyen)-[~]
$ fls -o 128 usb_fat_carving.001
r/r 3:  USB (Volume Label Entry)
d/d 6:  System Volume Information
r/r * 7:  _est
r/r 10: .dropbox.device
d/d * 13: old_File_Carving_files
r/r * 15: B_ub_poe4.bmp
r/r * 18: B_zoom-eubie-mono.bmp
r/r * 21: Ballardlab8.java
r/r * 24: brittLab10.java
r/r * 27: D0_example.doc
r/r * 30: D0_example2.doc
r/r * 33: G_BuiltForThis.gif
r/r * 35: G_zoom-sc.gif
r/r * 37: H_Form.html
r/r * 39: H_hello.html
r/r * 41: J_ub_law.jpg
r/r * 44: J_ub_night.jpg
r/r * 47: nps-2008-jean_outlook.pst
r/r * 50: P_CAS-zoom-6.png
r/r * 53: P_MSB_1_zoom.png
r/r * 57: pd_Evidence_search_techniques.pdf
r/r * 61: pd_Forensic_Report_Template.pdf
r/r * 64: pp_Number_Systems.pptx
r/r * 67: pp_one_page.pptx
r/r 69: readme.docx
r/r 70: readme.txt
r/r * 71: _tf_1.rtf
r/r * 74: T_Eubie-iphone-5-8.tiff
r/r * 78: T_youknowus-iphone-5-8.tiff
r/r * 81: W_axloadcomplete.wav
r/r * 84: W_speaker_test_sound.wav
r/r * 86: Z_file.zip
r/r * 88: 江莎莎行踪轨迹.doc
r/r * 91: nps-2008-jean_outlook.pst
v/v 3889667: $MBR
v/v 3889668: $FAT1
v/v 3889669: $FAT2
v/v 3889670: $OrphanFiles
```

- Decide which file types need to carve

```

hainguyen@hainguyen: ~
GNU nano 8.2 /etc/scalpel/scalpel.conf
# EXAMPLE WITH NO SUFFIX
#-----
#
# Here is an example of how to use the no extension option. Any files
# beginning with the string "FOREMOST" are carved and no file extensions
# are used. No footer is defined and the max carve size is 1000 bytes.
#
#     NONE      y      1000      FOREMOST
#-----
# GRAPHICS FILES
#-----
#
# AOL ART files
#     art      y      150000    \x4a\x47\x04\x0e      \xcf\xcf\xcb
#     art      y      150000    \x4a\x47\x03\x0e      \xd0\xcb\x00\x00
#
# GIF and JPG files (very common)
#     gif      y      5000000    \x47\x49\x46\x38\x37\x61      \x0>
#     gif      y      5000000    \x47\x49\x46\x38\x39\x61      \x0>
#     jpg      y      5242880    \xff\xd8\xff???Exif      \xf>
#     jpg      y      5242880    \xff\xd8\xff???JFIF      \xf>
#
# PNG
#     png      y      20000000    \x50\x4e\x47?      \xff\xfc\xfd\xfe
#
# BMP (used by MSWindows, use only if you have reason to think there are
# BMP files worth digging for. This often kicks back a lot of false
# positives
#     bmp      y      100000    BM??\x00\x00\x00
#
# TIFF
#     tif      y      200000000    \x49\x49\x2a\x00

```

- Save it and quit!
- Show help

```

(hainguyen@hainguyen)~$ scalpel -h
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.
Carves files from a disk image based on file headers and footers.

Usage: scalpel [-b] [-c <config file>] [-d] [-h|V] [-i <file>]
             [-m blocksize] [-n] [-o <outputdir>] [-O num] [-q clustersi
ze]
             [-r] [-s num] [-t <blockmap file>] [-u] [-v]
             <imgfile> [<imgfile>] ...

-b Carve files even if defined footers aren't discovered within
  maximum carve size for file type [foremost 0.69 compat mode].
-c Choose configuration file.
-d Generate header/footer database; will bypass certain optimizations
  and discover all footers, so performance suffers. Doesn't affect
  the set of files carved. **EXPERIMENTAL**
-h Print this help message and exit.
-i Read names of disk images from specified file.
-m Generate/update carve coverage blockmap file. The first 32bit
  unsigned int in the file identifies the block size. Thereafter
  each 32bit unsigned int entry in the blockmap file corresponds
  to one block in the image file. Each entry counts how many
  carved files contain this block. Requires more memory and
  disk. **EXPERIMENTAL**
-n Don't add extensions to extracted files.
-o Set output directory for carved files.
-O Don't organize carved files by type. Default is to organize carved files
  into subdirectories.
-p Perform image file preview; audit log indicates which files
  would have been carved, but no files are actually carved.
-q Carve only when header is cluster-aligned.
-r Find only first of overlapping headers/footers [foremost 0.69 compat mod
e].
-s Skip n bytes in each disk image before carving.
-t Set directory for coverage blockmap. **EXPERIMENTAL**
-u Use carve coverage blockmap when carving. Carve only sections
  of the image whose entries in the blockmap are 0. These areas

```

Step 3.

- Carving the USB image

```

(hainguyen@hainguyen)~$ scalpel usb_fat_carving.001 -o output
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/hainguyen/usb_fat_carving.001"

```

```

(hainguyen@hainguyen)~$ scalpel usb_fat_carving.001 -o output
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/hainguyen/usb_fat_carving.001"

Image file pass 1/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
gif with header "\x47\x49\x46\x38\x37\x61" and footer "\x00\x3b" --> 0 files
gif with header "\x47\x49\x46\x38\x39\x61" and footer "\x00\x3b" --> 0 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\x
\xd9" --> 8 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\x
\xd9" --> 9 files
jpg with header "\xff\xd8\xff\xe0\x00\x10\x4a\x46\x49\x46\x00\x01" and foote
r "\xff\xd9" --> 9 files
Carving files from image.
Image file pass 2/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 26, elapsed = 1 seconds.

```

- Show carved files

```

(hainguyen@hainguyen)~$ tree output
output
├── audit.txt
├── jpg-2-0
│   ├── 00000000.jpg
│   ├── 00000001.jpg
│   ├── 00000002.jpg
│   ├── 00000003.jpg
│   ├── 00000004.jpg
│   ├── 00000005.jpg
│   ├── 00000006.jpg
│   └── 00000007.jpg
├── jpg-3-0
│   ├── 00000008.jpg
│   ├── 00000009.jpg
│   ├── 00000010.jpg
│   ├── 00000011.jpg
│   ├── 00000012.jpg
│   ├── 00000013.jpg
│   ├── 00000014.jpg
│   ├── 00000015.jpg
│   └── 00000016.jpg
└── jpg-4-0
    ├── 00000017.jpg
    ├── 00000018.jpg
    ├── 00000019.jpg
    ├── 00000020.jpg
    ├── 00000021.jpg
    ├── 00000022.jpg
    ├── 00000023.jpg
    ├── 00000024.jpg
    └── 00000025.jpg

4 directories, 27 files

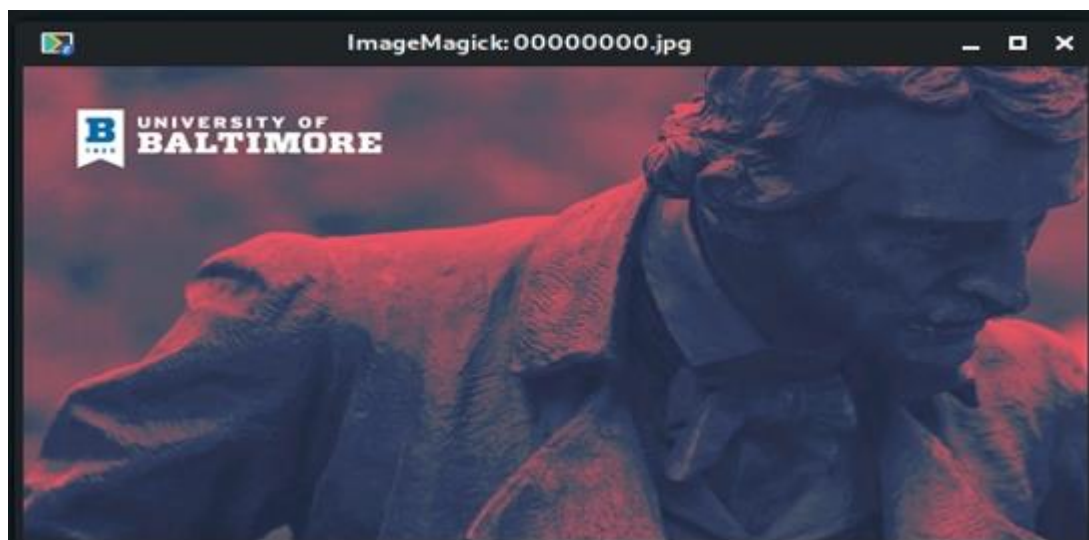
```

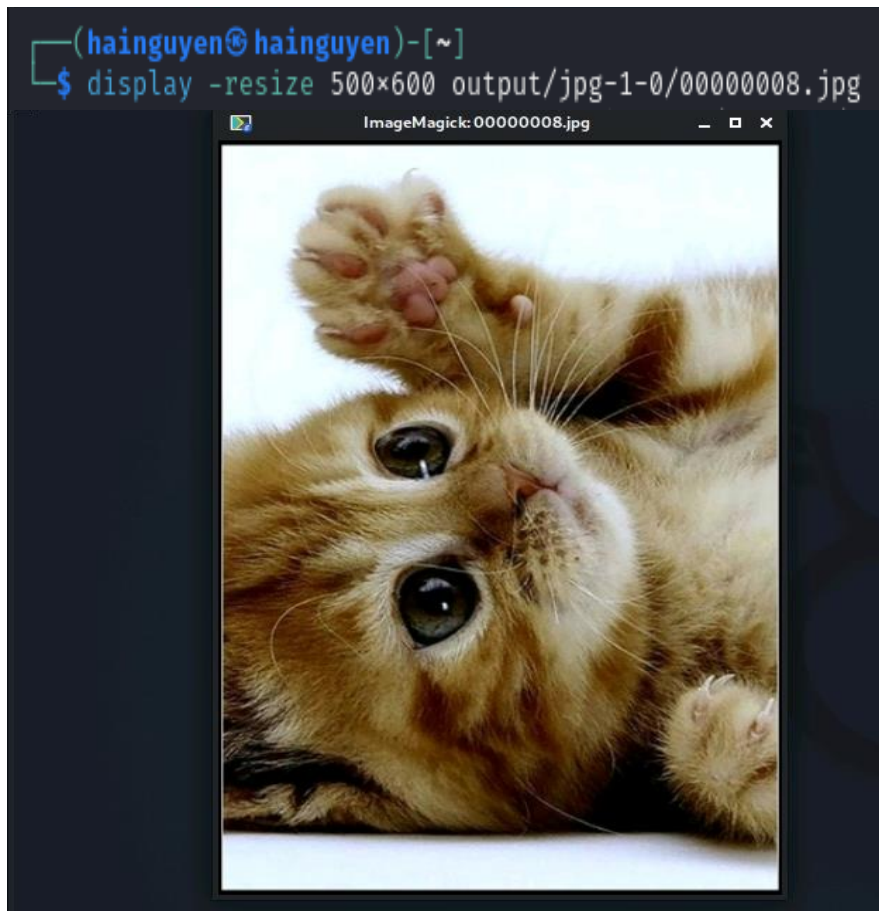
- Show audit log

```
hainguyen@hainguyen: ~  
└─(hainguyen@hainguyen)-[~]  
$ cat output/audit.txt  
  
Scalpel version 1.60 audit file  
Started at Thu May 22 12:26:16 2025  
Command line:  
scalpel usb_fat_carving.001 -o output  
  
Output directory: /home/hainguyen/output  
Configuration file: /etc/scalpel/scalpel.conf  
  
Opening target ~/home/hainguyen/usb_fat_carving.001*  
  
The following files were carved:  
File           Start           Chop           Length           Extracted From  
00000019.jpg   3796992         NO             99347            usb_fat_carving.001  
00000018.jpg   425822         NO             85046            usb_fat_carving.001  
00000017.jpg   335872         NO             85046            usb_fat_carving.001  
00000010.jpg   3796992         NO             3148500          usb_fat_carving.001  
00000009.jpg   425822         NO             4875802          usb_fat_carving.001  
00000008.jpg   335872         NO             4965752          usb_fat_carving.001  
00000002.jpg   6938624         NO             6868            usb_fat_carving.001  
00000001.jpg   5285888         NO             1659604          usb_fat_carving.001  
00000000.jpg   3897344         NO             3848148          usb_fat_carving.001  
00000004.jpg   15427584        NO             4223822          usb_fat_carving.001  
00000003.jpg   12881920        NO             4256310          usb_fat_carving.001  
00000006.jpg   19638272        NO             5229050          usb_fat_carving.001  
00000005.jpg   17121280        NO             4248530          usb_fat_carving.001  
00000007.jpg   21358592        NO             5183267          usb_fat_carving.001  
00000025.jpg   36671488        NO             292225          usb_fat_carving.001  
00000024.jpg   36571136        NO             99277            usb_fat_carving.001  
00000023.jpg   36393610        NO             2268            usb_fat_carving.001  
00000022.jpg   36352448        NO             2946            usb_fat_carving.001  
00000021.jpg   35508996        NO             13715           usb_fat_carving.001  
00000020.jpg   35350242        NO             239079          usb_fat_carving.001  
00000016.jpg   36671488        NO             2889262          usb_fat_carving.001  
00000015.jpg   36571136        NO             2989614          usb_fat_carving.001  
00000014.jpg   36393610        NO             3167140          usb_fat_carving.001  
00000013.jpg   36352448        NO             3208302          usb_fat_carving.001  
00000012.jpg   35508996        NO             3969754          usb_fat_carving.001  
00000011.jpg   35350242        NO             4210508          usb_fat_carving.001  
  
Completed at Thu May 22 12:26:17 2025
```

Step 4.

```
(hainguyen@hainguyen)-[~]  
$ display -resize 500x600 output/jpg-0-0/00000000.jpg
```





YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 2.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 2 From YOUR NAME**", replacing "YOUR NAME" with your real name.